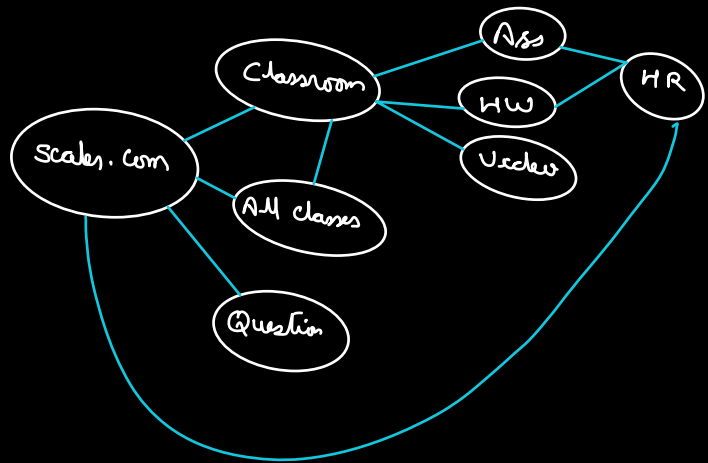
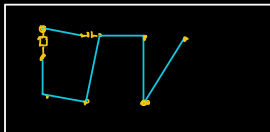
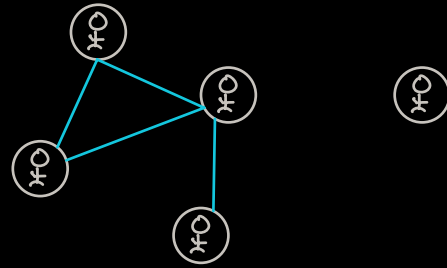
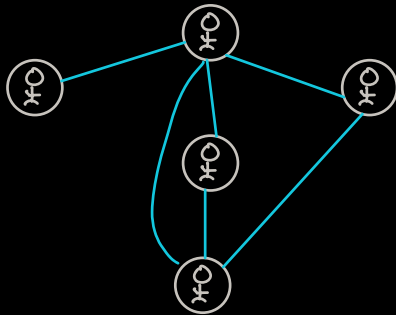
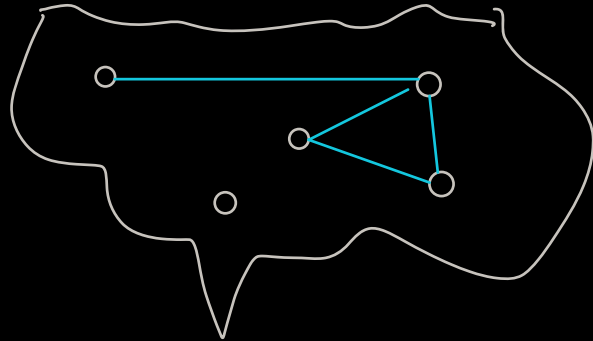


Graphs

A network of nodes & edges

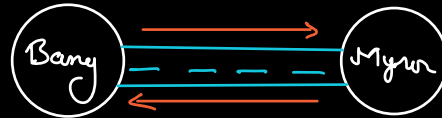


Facebook

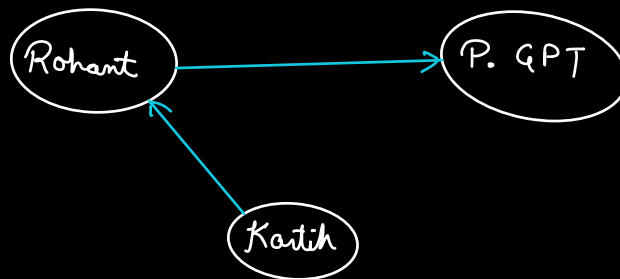


If Giriraj is a friend of Sahil
 $\Rightarrow$ that Sahil is also a friend of Giriraj

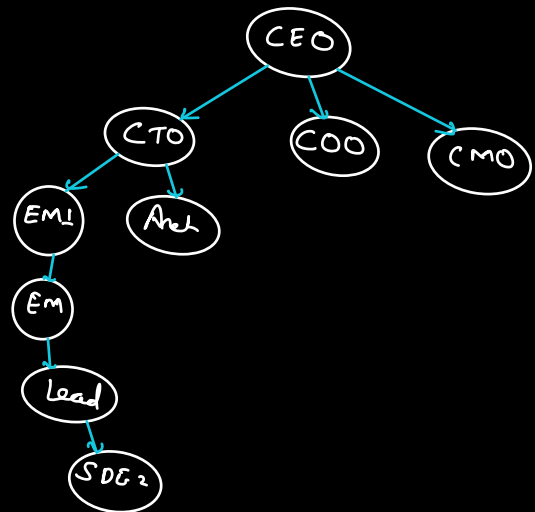
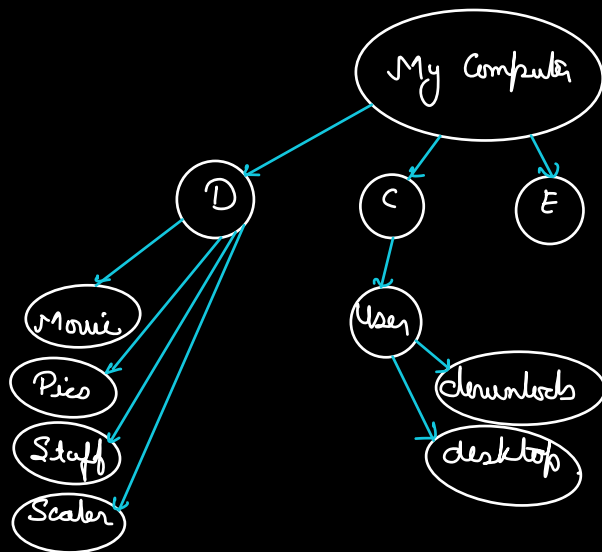
Undirected Graphs



Instagram
Twitter



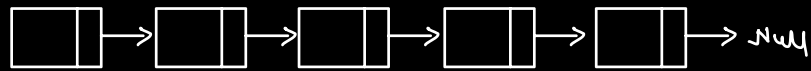
Directed Graphs



Tree

Graph which do not have
a cycle & do not have
unconnected nodes.





Tree & Graph

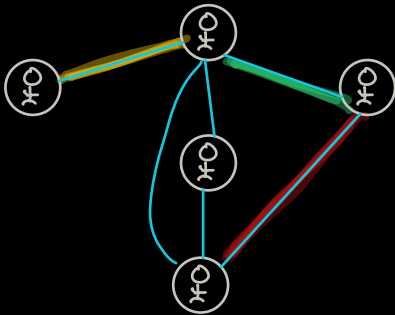
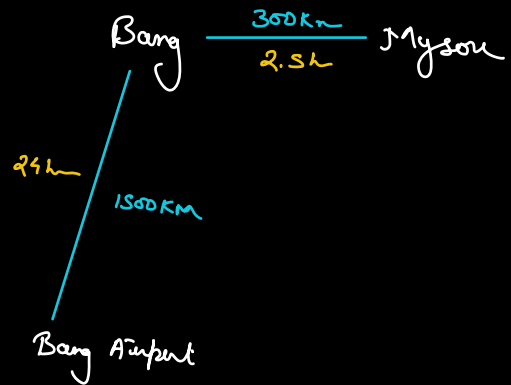


Graph but not a tree

Google Map

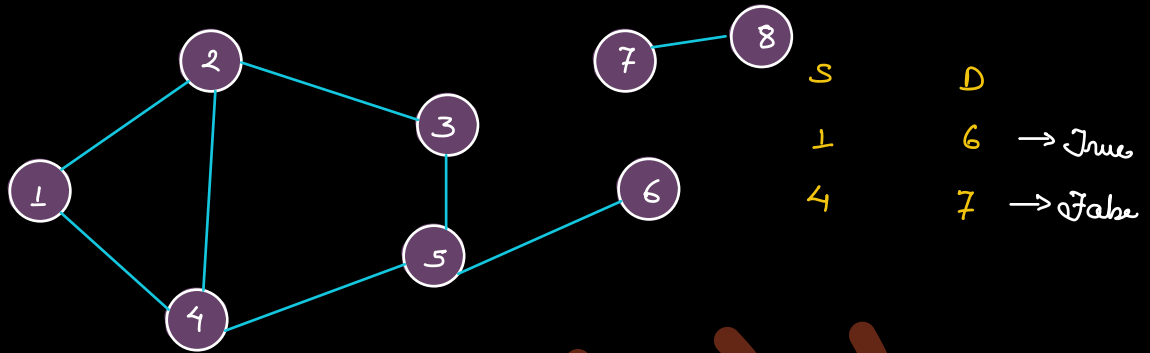
- distance
- traffic
- ...

Weighted Graphs



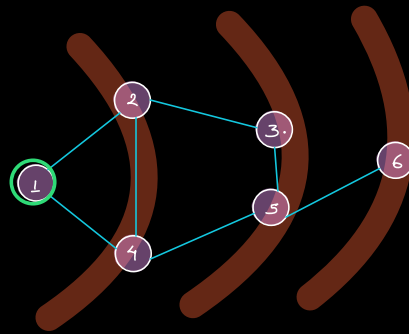
Cyclic

Q Given an undirected unweighted graph.
 Given a source node (S) & a destination node (D).
 Check if there exists a path from S to D.



Adj List

1 : {2, 4}
 2 : {1, 4, 3}
 3 : {2, 5}
 4 : {1, 2, 5}
 5 : {4, 3, 6}
 6 : {5}
 7 : {8}



Visited Set

1
 2
 4
 3
 5
 6

- Take a queue & a Set (visited nodes)
- Enqueue the source & mark as visited
- Till queue is not empty :
 - poll from queue $\rightarrow 4$
 - Iterate over all nodes connected to 4
 - for every node v in u .connections
 - if (v is not visited)

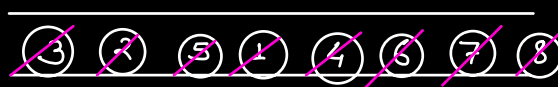
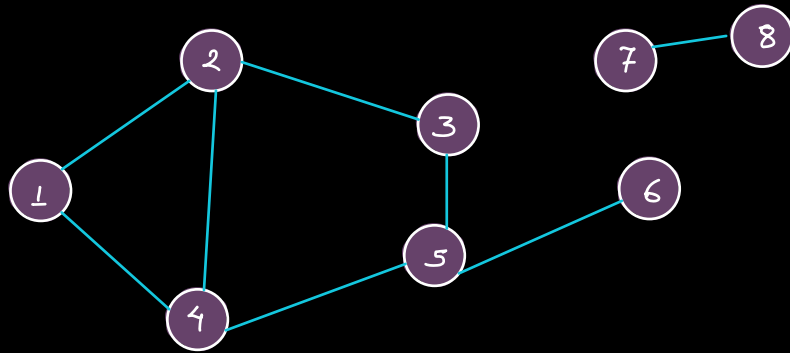
Tc:

Sc:

ret true if v is target
 mark visited
 & enqueue

Adj list

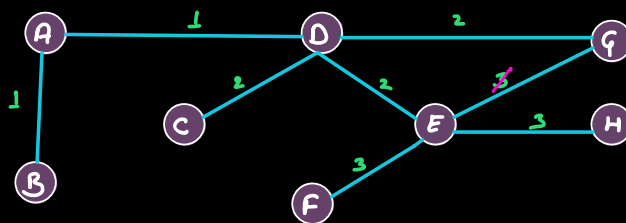
1 : {2, 4}
 2 : {1, 4, 3}
 3 : {2, 5}
 4 : {1, 2, 5}
 5 : {4, 3, 6}
 6 : {5}
 7 : {8}
 8 : {7}



Visited Set

3
 2
 5
 1
 4
 6
 7
 8

Q Given a connected undirected unweighted graph.
 & source & destination nodes.
 Find the length of shortest path from S to D.



S
A

D
H

→ 3

ADEH

~~{A, 0}~~ ~~{B, 1}~~ ~~{D, 1}~~ ~~{C, 2}~~ ~~{G, 2}~~ {E, 2} {F, 3} {H, 3}

Visited Set

A
 B
 D
 C
 E
 F
 H

BFS for shortest path →

works only for unweighted graphs
 or if all weights are same

- Q Given an undirected weighted graph with a S & D.
Find the min weighted path from S to D.
 $0 \leq \text{weight}(ij) \leq 2$

